Executive Report: IBM HR Attrition Analysis

1. Project Objective

The primary objective of this project is to analyze comprehensive employee data to identify the core factors contributing to employee attrition. By diagnosing these drivers, the organization can implement data-driven retention strategies and improve overall workforce stability.

2. Dataset Description

The analysis is based on the IBM HR Analytics dataset, which contains 1,470 unique employee records. Each record includes 35 distinct features (columns) providing a 360-degree view of the employee profile,

including:

- Demographics: Age, Gender, Education, Marital Status.
- Role & Compensation: Department, Job Role, Monthly Income, Stock Options.
- Work Dynamics: Overtime, Distance from Home, Business Travel frequency.
- Performance & Tenure: Years at Company, Job Satisfaction, Performance Rating.

3. Methodologies & Analysis Performed

To ensure data integrity and derive meaningful insights, the following analytical steps were executed:

- Data Loading & Cleaning: Ensuring no missing values or inconsistencies in the 1,470 records.
- Exploratory Data Analysis (EDA): Initial statistical overview of employee distribution.
- Attrition Correlation Analysis: Identifying the strongest links between work habits and exits.
- Departmental Deep-Dive: Comparative analysis of turnover rates across R&D, Sales, and HR.
- Overtime Impact Study: Quantifying the specific effect of extra hours on retention.
- Visualization: Utilizing Python libraries (Matplotlib/Seaborn) for high-impact visual storytelling.

4. Key Insights & Findings

- 16% Attrition Rate: 237 employees left the company, while 84% (1,233) stayed.
- Overtime Impact: Employees working overtime show a significantly higher attrition rate.
- R&D Concentration: Research & Development has the highest headcount and absolute attrition count.
- Sales Turnover: Noticeable employee turnover within Sales, indicating high-pressure dynamics.

- HR Stability: The Human Resources department maintains the lowest attrition rate in the company.

5. Conclusion & Recommendations

The analysis confirms that overtime and departmental pressure are critical drivers of turnover. Organizations should prioritize work-life balance and departmental workload management. Focusing on the R&D and Sales pipelines will yield the highest impact on overall retention.